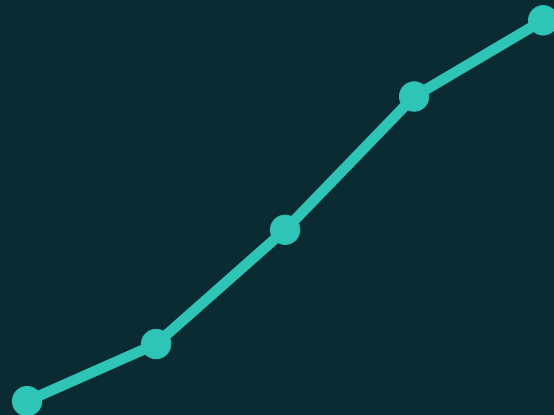


PROJECT PROPOSAL

BioTrack

Personal Health & Biomarker Analytics
Platform

Longitudinal trend analysis — not diagnosis.



Why BioTrack?



Scattered results

Lab reports sit in PDFs, portals and paper — never in one timeline.



Direction matters

A single value says little; the trajectory across months tells the story.



Numbers without context

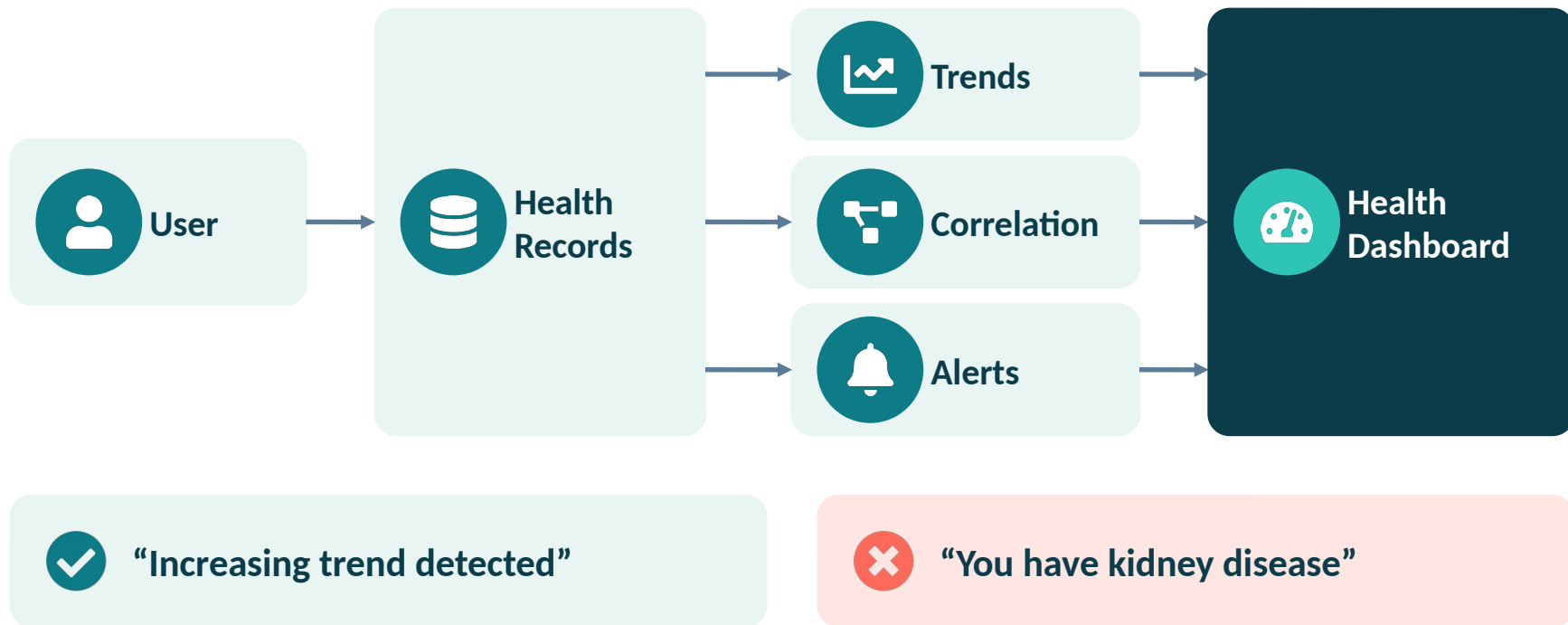
People see values but not trends, reference ranges or what changed.

OUR VISION

One place to record, trend and understand your biomarkers over time — in clear, non-diagnostic language.

Storing numbers is easy. The value is in longitudinal analysis.

Trends, not diagnoses



What users can track



Cardiovascular

Blood pressure · Heart rate · SpO₂ ·
Cholesterol



Metabolic

Weight · BMI · Blood glucose · HbA1c



Blood & nutrition

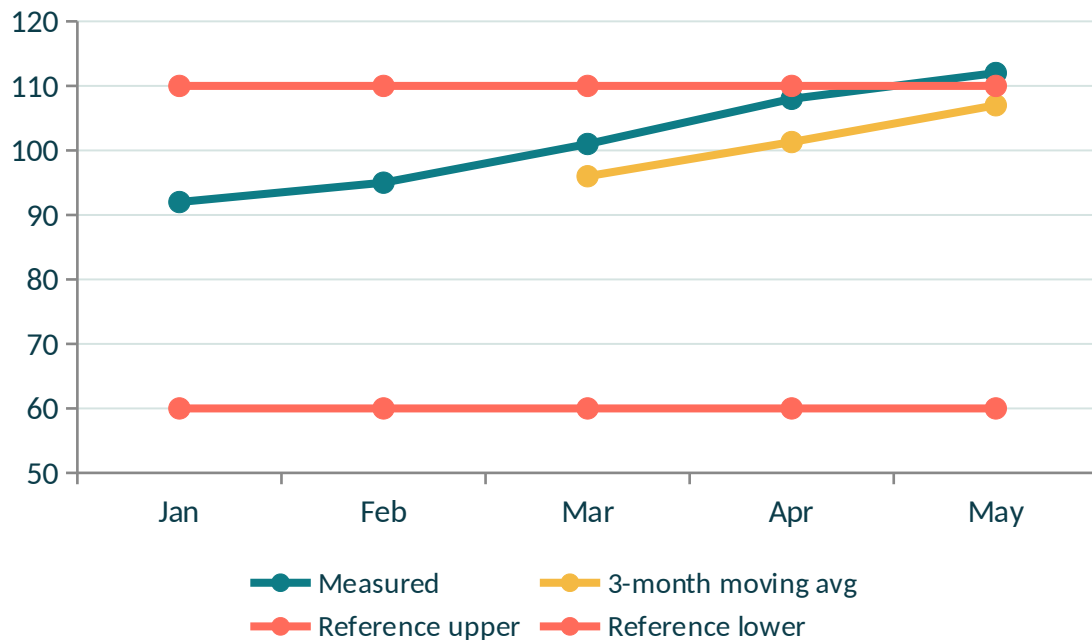
Hemoglobin · WBC · Vitamin D



Organ function

Creatinine · Liver enzymes

Example: creatinine over five months



Illustrative values in $\mu\text{mol/L}$. Reference limits vary by lab, age and sex.

+5.3

slope per month

+21.7%

change, Jan → May

107

3-month avg, May

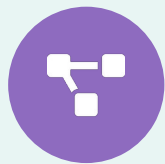
Increasing trend detected

Core features



Trend analysis

Slope, percentage change and moving averages for every biomarker.



Correlation

Explore how markers move together, e.g. weight and blood pressure.



Smart alerts

Out-of-range readings, rapid change and sustained drift notices.



Dashboard

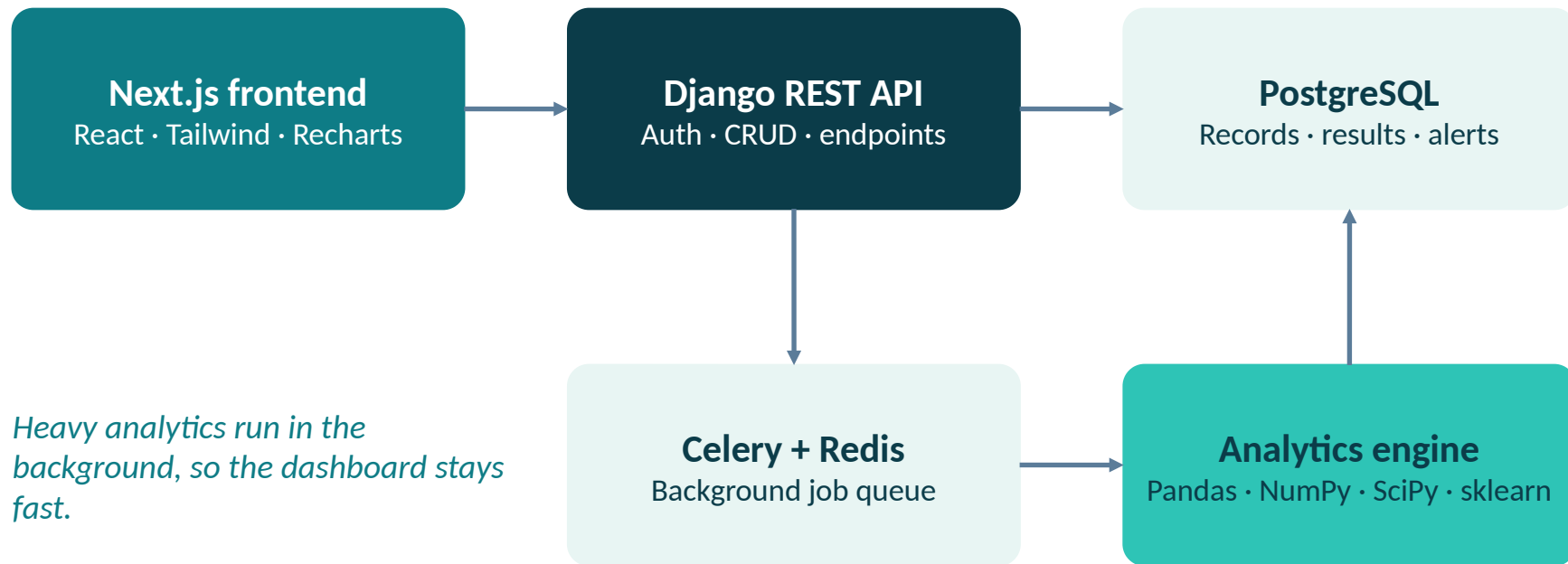
Interactive charts with reference-interval bands and CSV import.

How the analytics work

Technique	Tooling	Output
Linear regression	SciPy linregress	Slope and direction of change
% change, moving average	Pandas rolling	Smoothed series vs baseline
Trend significance	Kendall's tau (Mann-Kendall)	Is the trend real or noise?
Correlation	NumPy, SciPy	Pearson / Spearman across markers
Outlier detection	z-score, IsolationForest	Unusual readings flagged

Optional: short-horizon forecasts with scikit-learn, always shown with uncertainty.

System architecture



Technology stack



Frontend

- React / Next.js
- Tailwind CSS
- Recharts



Backend

- Django REST Framework
- PostgreSQL
- Redis / Celery



Analytics

- Python · Pandas
- NumPy · SciPy
- scikit-learn

Core data model

User

email · sex · birth year

Biomarker

name · unit · category

ReferenceRange

low · high · sex / age band

Measurement

user · biomarker · value ·
measured_at

TrendResult

slope · % change · moving avg

Alert

type · severity · message

Amber entities are derived from Measurements by the analytics engine.

Safety, privacy and ethics



BioTrack says

- Increasing trend detected
- Value outside reference range
- Consider discussing with a clinician



BioTrack never says

- “You have [disease]”
- “Start or stop this medication”
- “Your diagnosis is...”



Encrypted data, token-based auth



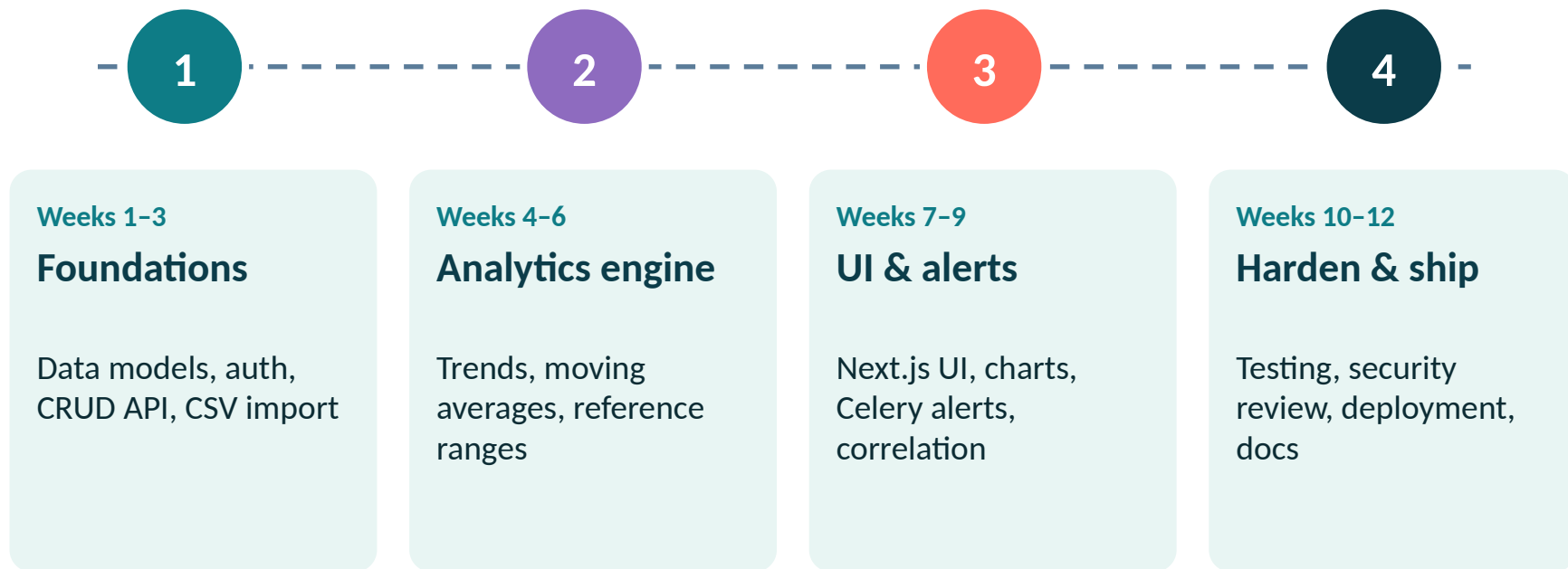
Export or delete data any time



Not a medical device; disclaimer on every view

Health data is sensitive: design for privacy regulations such as GDPR from day one.

Roadmap (12 weeks)



Outcomes and future scope

EXPECTED OUTCOMES

- Working MVP with trend dashboard
- Explainable, non-diagnostic analytics
- Clean API ready for new biomarkers

FUTURE SCOPE

- Wearable and health-app import
- Lab-report PDF extraction (OCR)
- Clinician sharing and mobile app

Thank you — questions?